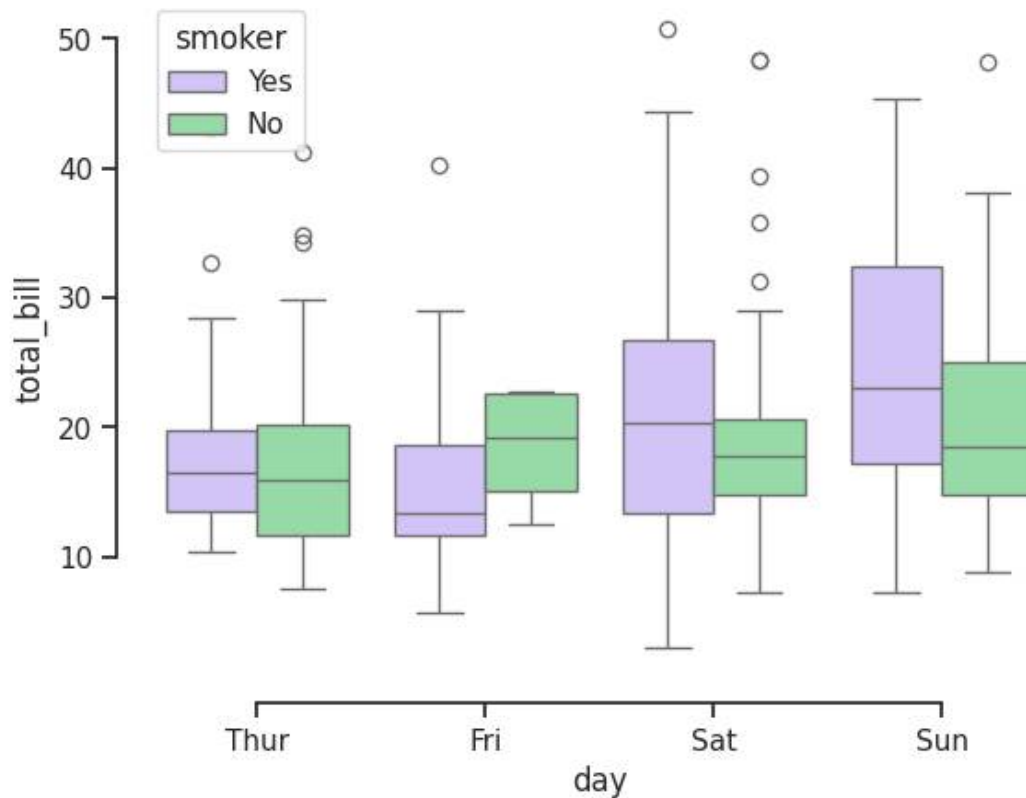


```
import seaborn as sns
# sns.set_theme(style="ticks", palette="pastel")

# Load the example tips dataset
tips = sns.load_dataset("tips")

# Draw a nested boxplot to show bills by day and time
sns.boxplot(x="day", y="total_bill",
            hue="smoker", palette=["m", "g"],
            data=tips)
sns.despine(offset=10, trim=True)
```



✓ EDA(Exploratory Data Analysis)

feature

tips

	total_bill	tip	sex	smoker	day	time	size	
0	16.99	1.01	Female	No	Sun	Dinner	2	
1	10.34	1.66	Male	No	Sun	Dinner	3	
2	21.01	3.50	Male	No	Sun	Dinner	3	
3	23.68	3.31	Male	No	Sun	Dinner	2	
4	24.59	3.61	Female	No	Sun	Dinner	4	
...	...	...	...	...	...	...	...	
239	29.03	5.92	Male	No	Sat	Dinner	3	
240	27.18	2.00	Female	Yes	Sat	Dinner	2	
241	22.67	2.00	Male	Yes	Sat	Dinner	2	
242	17.82	1.75	Male	No	Sat	Dinner	2	
243	18.78	3.00	Female	No	Thur	Dinner	2	

244 rows × 7 columns

다음 단계:

[tips 변수로 코드 생성](#)[New interactive sheet](#)

tips.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 244 entries, 0 to 243
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype
---  -
0   total_bill  244 non-null    float64
1   tip         244 non-null    float64
2   sex         244 non-null    category
3   smoker      244 non-null    category
4   day         244 non-null    category
5   time        244 non-null    category
6   size        244 non-null    int64
dtypes: category(4), float64(2), int64(1)
memory usage: 7.4 KB
```

```
# df = tips
df = tips.copy()
```

%whos

Variable	Type	Data/Info
df	DataFrame	total_bill tip <...>nWn[244 rows x 7 columns]
df_filtered	DataFrame	total_bill tip <...>WnWn[81 rows x 7 columns]
path	str	/content/sample_data/california_housing_train.csv
pd	module	<module 'pandas' from '/u<...>ages/pandas/__init__.py'>
plt	module	<module 'matplotlib.pyplot' from '/es/matplotlib/pyplot.py'>
result	DataFrame	total_bill tip s<...>t Dinner 1 32.57329
scatter	PathCollection	<matplotlib.collections.PathCollection object at 0x7e2fa1296d50>
sns	module	<module 'seaborn' from '/<...>ges/seaborn/__init__.py'>
tips	DataFrame	total_bill tip <...>nWn[244 rows x 7 columns]

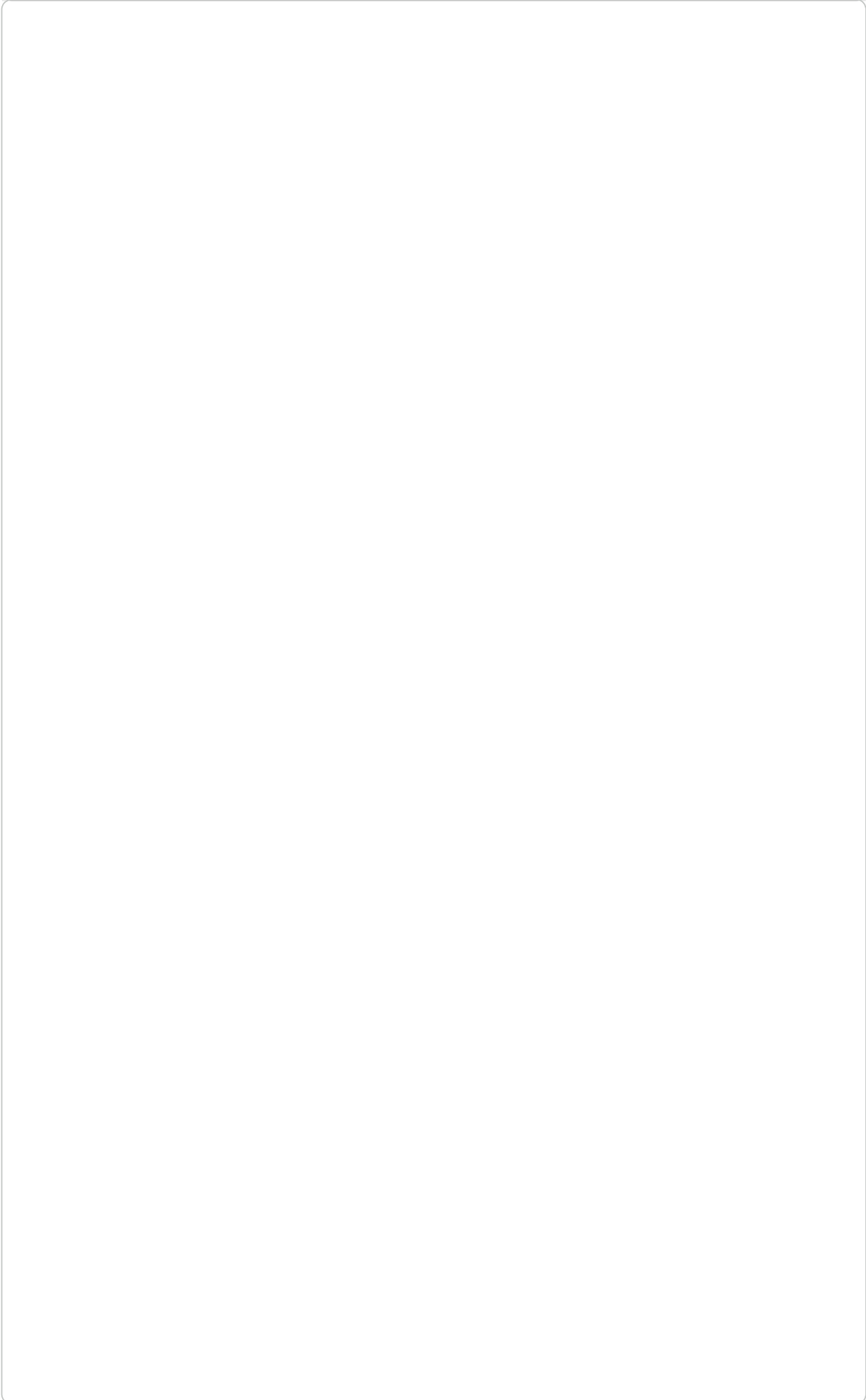
```
(df['tip'] / df['total_bill']) * 100
```

```
0
0    5.944673
1    16.054159
2    16.658734
3    13.978041
4    14.680765
...
239  20.392697
240   7.358352
241   8.822232
242   9.820426
243  15.974441
```

244 rows × 1 columns

dtype: float64

```
df['tip_pct'] = (df['tip'] / df['total_bill']) * 100
df[(df['tip_pct'] >= 20)]
```



	total_bill	tip	sex	smoker	day	time	size	tip_pct	
6	8.77	2.00	Male	No	Sun	Dinner	2	22.805017	
9	14.78	3.23	Male	No	Sun	Dinner	2	21.853857	
14	14.83	3.02	Female	No	Sun	Dinner	2	20.364127	

tips

18	16.97	3.50	Female	No	Sun	Dinner	3	20.724632	
20	17.02	4.00	Male	No	Sat	Dinner	2	22.167857	
0	16.99	1.01	Female	No	Sun	Dinner	2		
42	15.34	3.00	Male	No	Sun	Dinner	2	21.551220	
1	10.34	1.66	Male	No	Sun	Dinner	3		
16	22.22	5.00	Male	No	Sun	Dinner	2	22.492128	
2	21.01	3.50	Male	No	Sun	Dinner	3		
31	10.29	2.00	Female	No	Sun	Dinner	2	25.267250	
3	23.68	3.31	Male	No	Sun	Dinner	2		
62	18.28	2.76	Male	Yes	Sat	Dinner	4	20.557682	
4	24.59	3.61	Female	No	Sun	Dinner	4		
81	3.07	1.00	Female	Yes	Sat	Dinner	1	32.573290	
81	16.66	2.40	Male	No	Thur	Lunch	2	20.408163	
239	29.03	5.92	Male	No	Sat	Dinner	3		
81	18.28	4.00	Male	No	Thur	Lunch	2	21.881838	
240	27.18	2.00	Female	Yes	Sat	Dinner	2		
88	24.71	5.85	Male	No	Thur	Lunch	2	23.674626	
241	22.67	2.00	Male	Yes	Sat	Dinner	2		
33	10.32	4.30	Female	Yes	Fri	Dinner	2	26.348039	
242	17.82	1.75	Male	No	Sat	Dinner	2		
100	11.25	2.50	Female	Yes	Fri	Dinner	2	22.026432	
243	18.78	3.00	Female	No	Thur	Dinner	2		
100	18.24	3.10	Male	No	Sat	Dinner	2	20.614035	

244 rows × 7 columns

다음 단계: tips 변수의 코드로 생성 [New interactive sheet](#)

115	17.31	3.50	Female	No	Sun	Dinner	2	20.219526	
-----	-------	------	--------	----	-----	--------	---	-----------	--

id(df)

138743277212672	13.16	2.75	Female	No	Thur	Lunch	2	20.896657	
-----------------	-------	------	--------	----	------	-------	---	-----------	--

id(tips)

149	7.51	2.00	Male	No	Thur	Lunch	2	28.031158	
138743031082608									
172	7.25	5.15	Male	Yes	Sun	Dinner	2	71.034483	

174 Prompt: 이 데이터 범주형과 수치형 데이터의 변수를 분석하는 시각화와 두

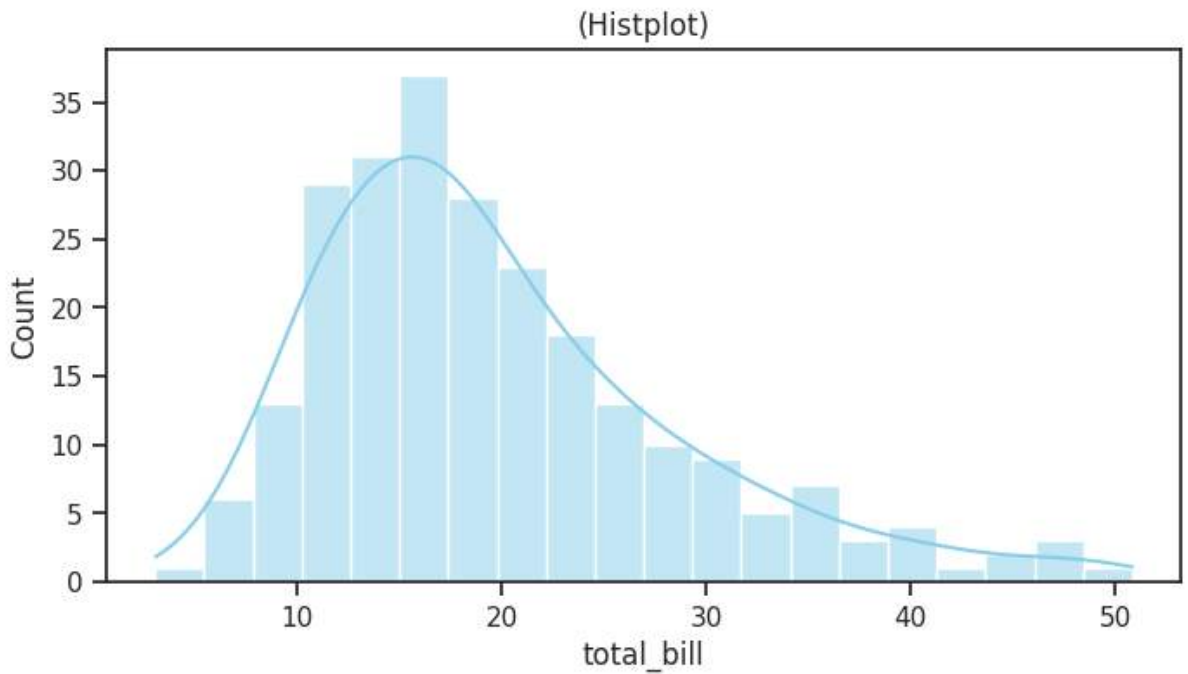
178개, 두개 이상에 대한 열 분석 시각화를 seaborn 코드로 주라. 2 41.666667

181	23.33	5.65	Male	Yes	Sun	Dinner	2	24.217745	
-----	-------	------	------	-----	-----	--------	---	-----------	--

```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
plt.figure(figsize=(8, 4))
# total_bill의 분포 확인 (kde=True로 부드러운 곡선 추가)
sns.histplot(data=tips, x="total_bill", kde=True, bins=20, color="skyblue")
plt.title("(Histplot)")
plt.show()
```

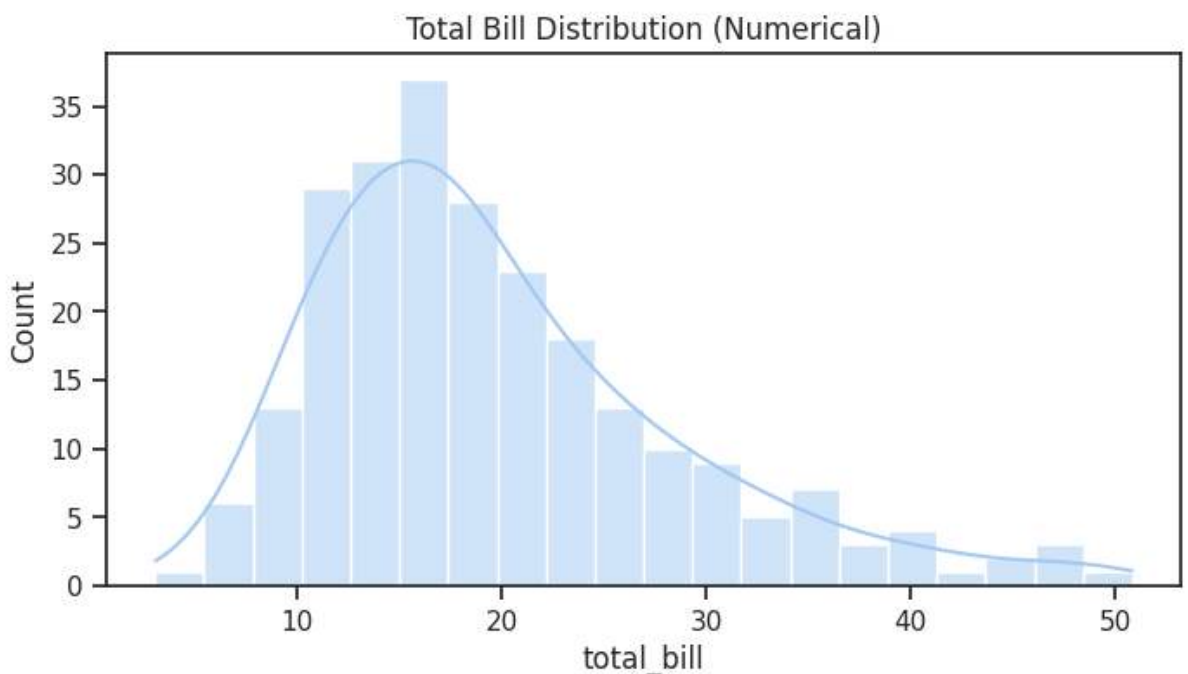
221	13.42	3.48	Female	Yes	Fri	Lunch	2	25.931446	
-----	-------	------	--------	-----	-----	-------	---	-----------	--



✓ (1) 수치형 변수 단변량: total_bill 분포 (Histogram + KDE)

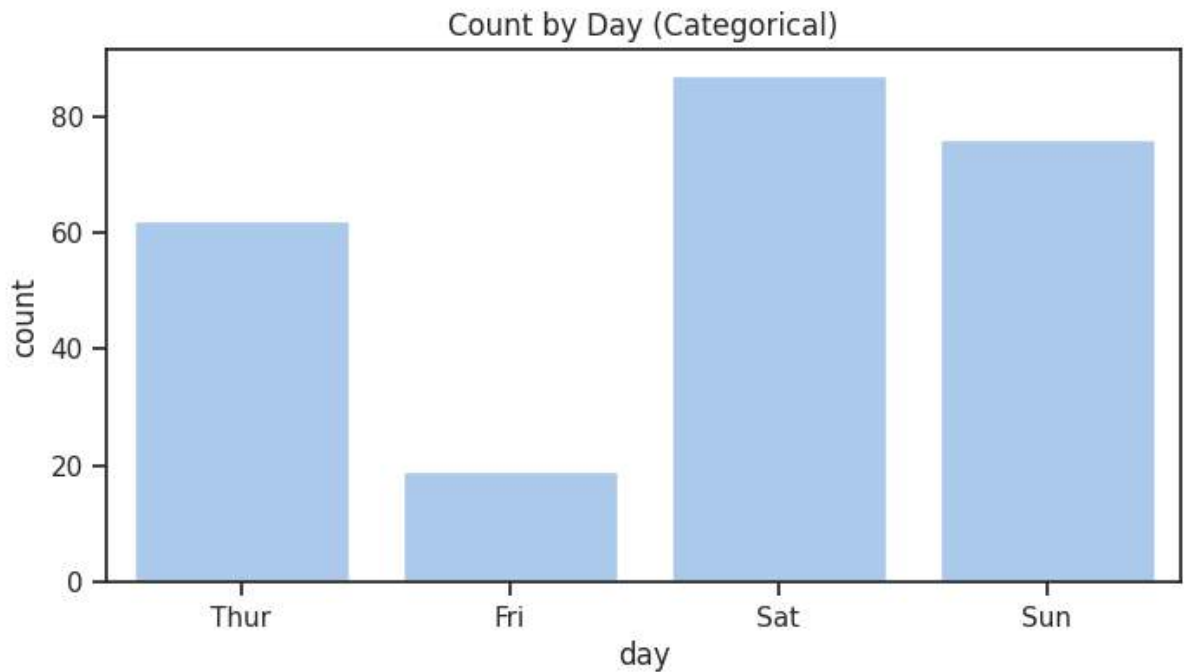
```
# @title (1) 수치형 변수 단변량: total_bill 분포 (Histogram + KDE)
```

```
import matplotlib.pyplot as plt
plt.figure(figsize=(8, 4))
sns.histplot(data=df, x='total_bill', kde=True, bins=20)
plt.title("Total Bill Distribution (Numerical)")
plt.show()
```



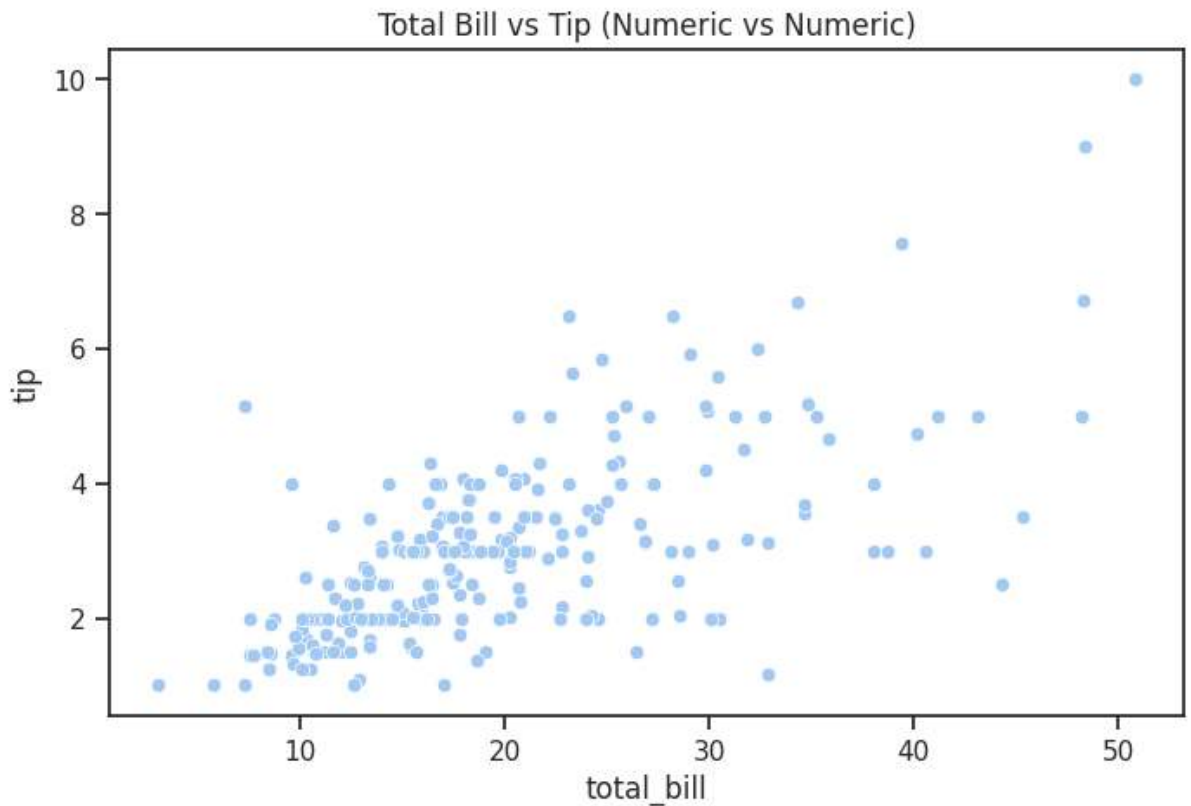
✓ (2) 범주형 변수 단변량: day 요일별 방문 빈도 (Count Plot)

```
# @title (2) 범주형 변수 단변량: day 요일별 방문 빈도 (Count Plot)
plt.figure(figsize=(8, 4))
sns.countplot(data=df, x='day')
plt.title("Count by Day (Categorical)")
plt.show()
```



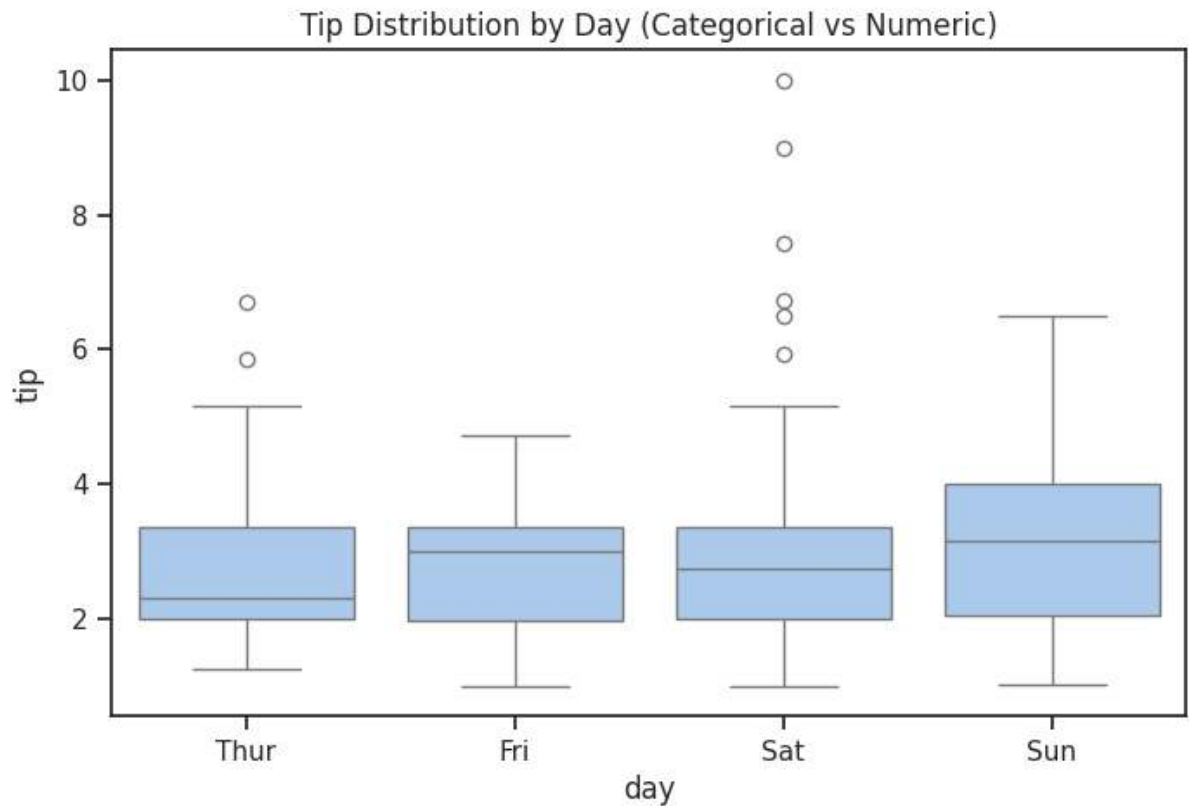
✓ (1) 수치형 vs 수치형: total_bill과 tip의 관계 (Scatter Plot)

```
# @title (1) 수치형 vs 수치형: total_bill과 tip의 관계 (Scatter Plot)
plt.figure(figsize=(8, 5))
sns.scatterplot(data=df, x='total_bill', y='tip')
plt.title("Total Bill vs Tip (Numeric vs Numeric)")
plt.show()
```



✓ (2) 범주형 vs 수치형: day별 tip 분포 비교 (Box Plot)

```
# @title (2) 범주형 vs 수치형: day별 tip 분포 비교 (Box Plot)
plt.figure(figsize=(8, 5))
sns.boxplot(data=df, x='day', y='tip')
plt.title("Tip Distribution by Day (Categorical vs Numeric)")
plt.show()
```



```
df.head()
```

	total_bill	tip	sex	smoker	day	time	size	tip_pct	
0	16.99	1.01	Female	No	Sun	Dinner	2	5.944673	
1	10.34	1.66	Male	No	Sun	Dinner	3	16.054159	
2	21.01	3.50	Male	No	Sun	Dinner	3	16.658734	
3	23.68	3.31	Male	No	Sun	Dinner	2	13.978041	
4	24.59	3.61	Female	No	Sun	Dinner	4	14.680765	

다음 단계:

[df 변수로 코드 생성](#)[New interactive sheet](#)

```
df.groupby('day')['tip'].describe()
```

```
/tmp/ipykernel_1563/3048255687.py:1: FutureWarning: The default of observed=False is deprecated
df.groupby('day')['tip'].describe()
```

	count	mean	std	min	25%	50%	75%	max	
day									
Thur	62.0	2.771452	1.240223	1.25	2.0000	2.305	3.3625	6.70	
Fri	19.0	2.734737	1.019577	1.00	1.9600	3.000	3.3650	4.73	
Sat	87.0	2.993103	1.631014	1.00	2.0000	2.750	3.3700	10.00	
Sun	76.0	3.255132	1.234880	1.01	2.0375	3.150	4.0000	6.50	

```
df.groupby('day').describe()
```

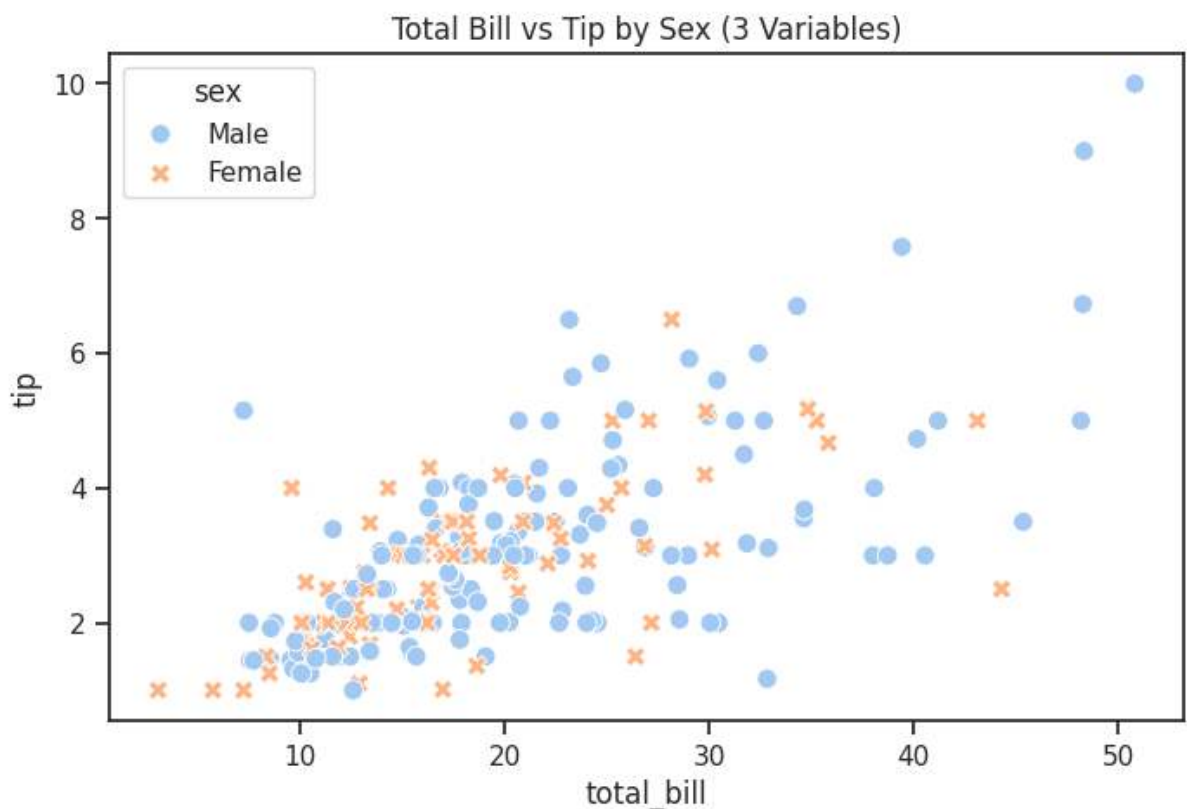
```
/tmp/ipykernel_1563/1891411268.py:1: FutureWarning: The default of observed=False is deprecated
df.groupby('day').describe()
```

	total_bill								tip	
	count	mean	std	min	25%	50%	75%	max	count	mean
day										
Thur	62.0	17.682742	7.886170	7.51	12.4425	16.20	20.1550	43.11	62.0	2.771452
Fri	19.0	17.151579	8.302660	5.75	12.0950	15.38	21.7500	40.17	19.0	2.734737
Sat	87.0	20.441379	9.480419	3.07	13.9050	18.24	24.7400	50.81	87.0	2.993103
Sun	76.0	21.410000	8.832122	7.25	14.9875	19.63	25.5975	48.17	76.0	3.255132

4 rows × 32 columns

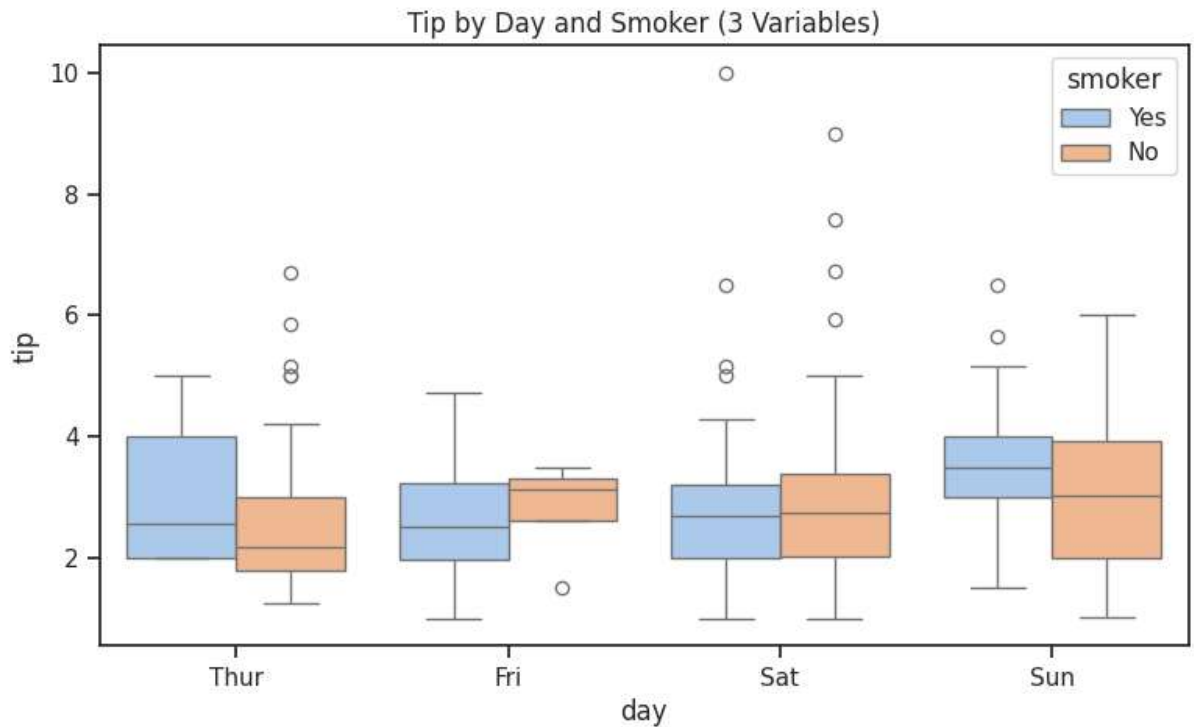
✓ (1) 수치형 2개 + 범주형 1개 (hue 활용): total_bill과 tip 관계를 sex별로 구분

```
# @title (1) 수치형 2개 + 범주형 1개 (hue 활용): total_bill과 tip 관계를 sex별로 구분
plt.figure(figsize=(8, 5))
sns.scatterplot(data=df, x='total_bill', y='tip', hue='sex', style='sex', s=70)
plt.title("Total Bill vs Tip by Sex (3 Variables)")
plt.show()
```



✓ (2) 범주형 2개 + 수치형 1개: day 및 smoker별 tip 중앙값/분포 비교 (Catplot / Boxplot)

```
# @title (2) 범주형 2개 + 수치형 1개: day 및 smoker별 tip 중앙값/분포 비교 (Catplot / Boxplot)
plt.figure(figsize=(9, 5))
sns.boxplot(data=df, x='day', y='tip', hue='smoker')
plt.title("Tip by Day and Smoker (3 Variables)")
plt.show()
```



```
path = "/content/sample_data/california_housing_train.csv"
```

```
import pandas as pd
df = pd.read_csv(path)
df.columns
```

```
Index(['longitude', 'latitude', 'housing_median_age', 'total_rooms',
       'total_bedrooms', 'population', 'households', 'median_income',
       'median_house_value'],
      dtype='object')
```

지도를 그려서 median_house_value/population 별로 scatter하여 어느 지역의 집 값이 비싼지 말해줘. 시각화도 해줘.

```
import matplotlib.pyplot as plt

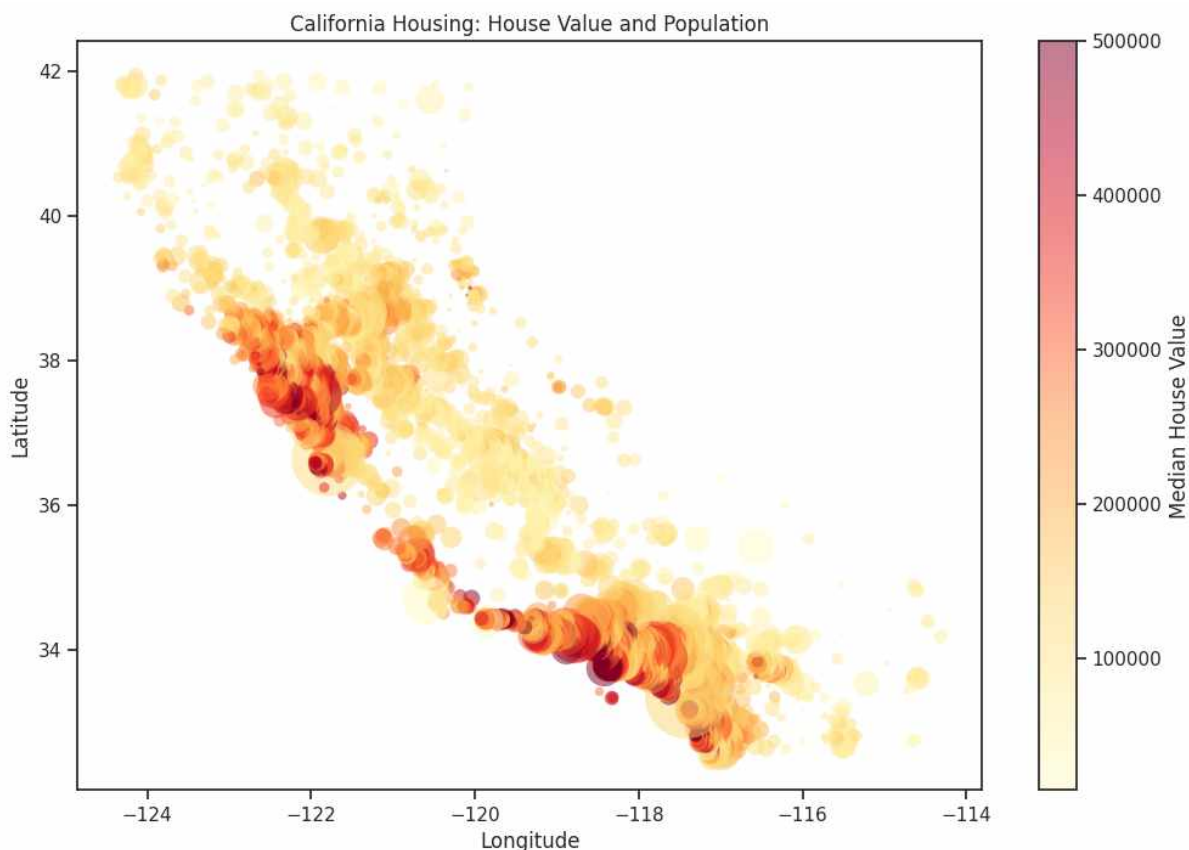
plt.figure(figsize=(12, 8))

scatter = plt.scatter(
    df['longitude'],          # x축: 경도
    df['latitude'],          # y축: 위도
    c=df['median_house_value'], # 색깔: 집값
    s=df['population'] / 15,  # 점 크기: 인구
    cmap='YlOrRd',           # 노란색 → 빨간색
    alpha=0.5,
    edgecolors='none'
)
```

```
# 색깔 막대
plt.colorbar(
    scatter,
    label='Median House Value'
)

# 그래프 제목 및 축 이름
plt.xlabel('Longitude')
plt.ylabel('Latitude')
plt.title('California Housing: House Value and Population')

plt.show()
```



```
df.columns
```

```
Index(['longitude', 'latitude', 'housing_median_age', 'total_rooms',
       'total_bedrooms', 'population', 'households', 'median_income',
       'median_house_value'],
      dtype='object')
```

```
df = tips.copy()
```

```
# -----
# [방법 1] 단일 조건별 평균 Tip 금액 분석
# -----
print("=== 요일별 평균 Tip ===")
print(df.groupby('day')['tip'].mean().sort_values(ascending=False))

print("\n=== 시간대별 평균 Tip ===")
print(df.groupby('time')['tip'].mean().sort_values(ascending=False))
```

```
print("\n=== 성별별 평균 Tip ===")
print(df.groupby('sex')['tip'].mean().sort_values(ascending=False))

print("\n=== 흡연 여부별 평균 Tip ===")
print(df.groupby('smoker')['tip'].mean().sort_values(ascending=False))

print("\n=== 식사 인원(size)별 평균 Tip ===")
print(df.groupby('size')['tip'].mean().sort_values(ascending=False))
```

```
=== 요일별 평균 Tip ===
day
Sun      3.255132
Sat      2.993103
Thur     2.771452
Fri      2.734737
Name: tip, dtype: float64
```

```
=== 시간대별 평균 Tip ===
time
Dinner   3.102670
Lunch    2.728088
Name: tip, dtype: float64
```

```
=== 성별별 평균 Tip ===
sex
Male     3.089618
Female   2.833448
Name: tip, dtype: float64
```

```
=== 흡연 여부별 평균 Tip ===
smoker
Yes      3.008710
No       2.991854
Name: tip, dtype: float64
```

```
=== 식사 인원(size)별 평균 Tip ===
size
6      5.225000
4      4.135405
5      4.028000
3      3.393158
2      2.582308
1      1.437500
Name: tip, dtype: float64
```

```
/tmp/ipykernel_1563/3391474988.py:5: FutureWarning: The default of observed=False is deprecated
print(df.groupby('day')['tip'].mean().sort_values(ascending=False))
/tmp/ipykernel_1563/3391474988.py:8: FutureWarning: The default of observed=False is deprecated
print(df.groupby('time')['tip'].mean().sort_values(ascending=False))
/tmp/ipykernel_1563/3391474988.py:11: FutureWarning: The default of observed=False is deprecated
print(df.groupby('sex')['tip'].mean().sort_values(ascending=False))
/tmp/ipykernel_1563/3391474988.py:14: FutureWarning: The default of observed=False is deprecated
print(df.groupby('smoker')['tip'].mean().sort_values(ascending=False))
```

```
import matplotlib.pyplot as plt

# Filter data for Thursday and Friday
df_filtered = df[df['day'].isin(['Thur', 'Fri'])]

# Create a box plot to visualize the tip distribution for 'Thur' and 'Fri'
plt.figure(figsize=(8, 6))
sns.boxplot(x='day', y='tip', data=df_filtered, palette=['skyblue', 'lightcoral'])
```

```
plt.title('Tip Distribution for Thursday and Friday')
plt.xlabel('Day of the Week')
plt.ylabel('Tip Amount')
sns.despine(offset=10, trim=True)
plt.show()
```

/tmp/ipykernel_1563/401968391.py:8: FutureWarning:

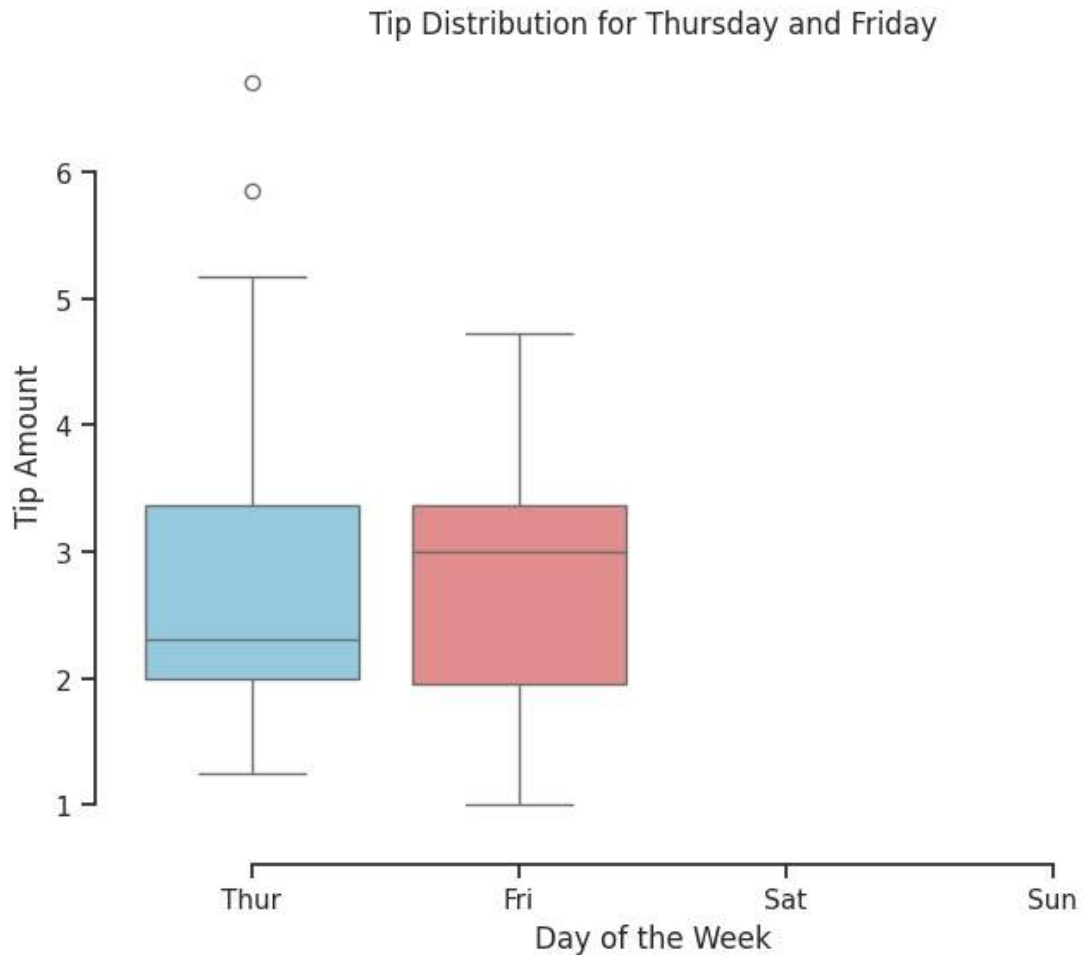
Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign

```
sns.boxplot(x='day', y='tip', data=df_filtered, palette=['skyblue', 'lightcoral'])
```

/tmp/ipykernel_1563/401968391.py:8: UserWarning:

The palette list has fewer values (2) than needed (4) and will cycle, which may produce an un

```
sns.boxplot(x='day', y='tip', data=df_filtered, palette=['skyblue', 'lightcoral'])
```



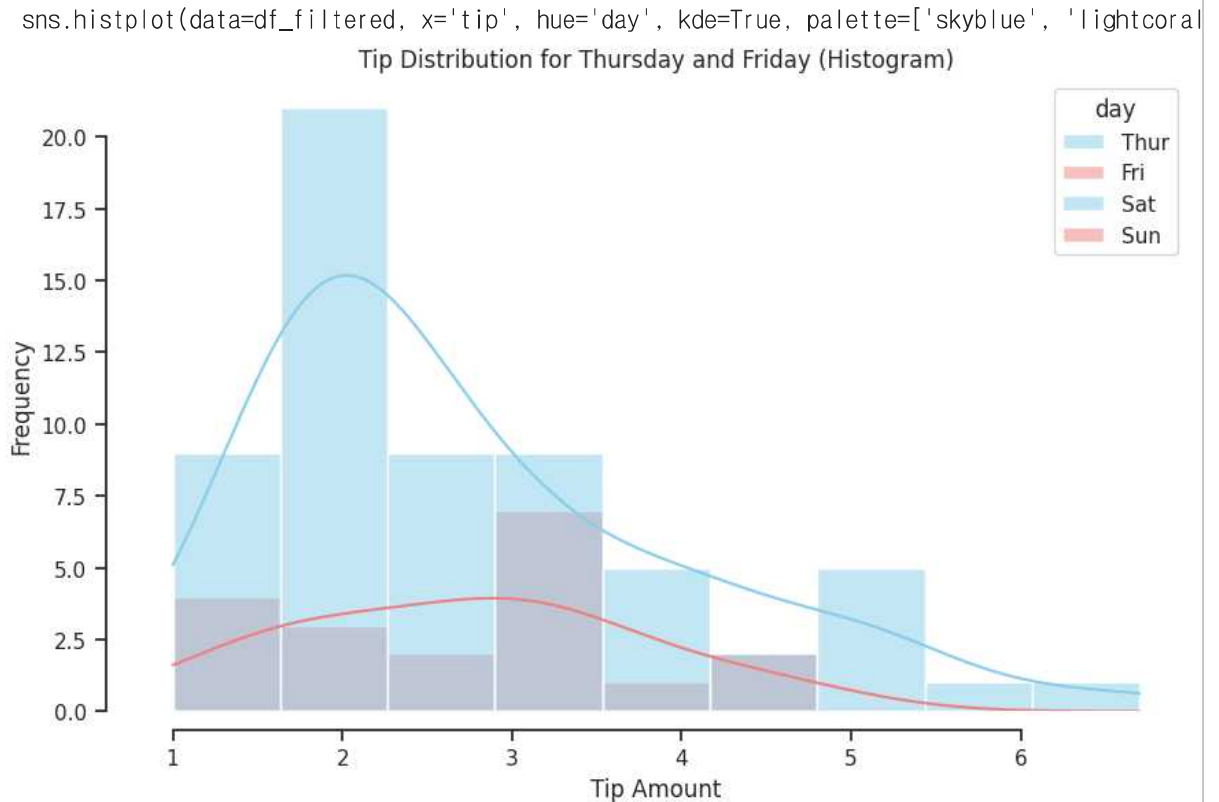
```
import seaborn as sns
import matplotlib.pyplot as plt

# Filter data for Thursday and Friday (df_filtered is already available)
df_filtered = df[df['day'].isin(['Thur', 'Fri'])]

plt.figure(figsize=(10, 6))
sns.histplot(data=df_filtered, x='tip', hue='day', kde=True, palette=['skyblue', 'lightcoral'])
plt.title('Tip Distribution for Thursday and Friday (Histogram)')
plt.xlabel('Tip Amount')
plt.ylabel('Frequency')
sns.despine(offset=10, trim=True)
plt.show()
```

/tmp/ipykernel_1563/2219589828.py:8: UserWarning:

The palette list has fewer values (2) than needed (4) and will cycle, which may produce an undesirable color palette.



Questions : 가장 많은 Tip을 받으려면 어떤 요일, 시간대? 어떤 유형의 손님에게 하여야 하나?

요일은 SAT에 방문할 것이다. 팁 평균은 낮지만 이 날에 방문 빈도가 높고(높다는 것은 팁을 받을 회수가 늘어난다고 생각함.) 이상치가 높아서 운이 좋으면 큰 금액을 받을 수 있다. 하지만 오랫동안 알바를 한다면 평균치에 수렴할 것이기 때문에 가장 많은 팁을 받으려면 고객 방문 빈도가 중요하다고 생각한다.

그리고 저녁에 방문할 것이다. SAT 날에는 사람들이 디너만 온다.

```
plt.figure(figsize=(8, 4))
sns.countplot(data=df, x='time')
plt.title("Count by Day (Categorical)")
plt.show()
```

